

Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. A developer fixes a bug in a specific module, and the tester verifies the fix. Following this, the tester runs a suite of existing test cases across the entire application to ensure no new defects were introduced. What is this secondary activity called? (1 Marks)

- A. Re-testing
- B. Regression testing
- C. Sanity testing
- D. Integration testing

Q2. An input field accepts integers from 10 to 50 inclusive. Using Boundary Value Analysis (BVA), which set of values represents the most effective test cases for the boundaries? (1 Marks)

- A. 9, 10, 50, 51
- B. 10, 30, 50
- C. 9, 10, 11, 49, 50, 51
- D. 0, 10, 50, 100

Q3. A system calculates discounts based on age: under 18 (10% discount), 18 to 60 (no discount), and over 60 (15% discount). What is the minimum number of test cases required to cover all valid equivalence partitions? (1 Marks)

- A. 2
- B. 3
- C. 6
- D. 4

Q4. During integration testing, a team uses 'stubs' to simulate the behavior of lower-level modules that are not yet developed. Which integration testing strategy is being employed? (1 Marks)

- A. Bottom-up integration
- B. Top-down integration
- C. Big Bang integration
- D. Sandwich integration

Q5. A testing team is evaluating how a web application behaves when 10,000 concurrent users access it simultaneously over an extended period to identify memory leaks and resource degradation. What type of testing is this? (1 Marks)

- A. Load testing
- B. Stress testing
- C. Endurance testing

D. Volume testing

Q6. A tester is analyzing the internal control flow and path coverage of a Java method to design test cases. What category of testing is this tester performing? (1 Marks)

- A. Black-box testing
- B. White-box testing
- C. Gray-box testing
- D. System testing

Q7. A software company releases a pre-production version of its mobile app to a selected group of external users in the target market to gather feedback in real-world environments. What is this testing phase called? (1 Marks)

- A. Alpha testing
- B. Beta testing
- C. System integration testing
- D. User acceptance testing (UAT)

Q8. An experienced tester is given a new application with no formal requirements documentation. They simultaneously design, execute, and learn about the system on the fly. What testing approach is this? (1 Marks)

- A. Scripted testing
- B. Exploratory testing
- C. Ad-hoc testing
- D. Sanity testing

Q9. A QA team intentionally introduces small, single code changes (faults) into a program to evaluate whether their existing test suite can detect and fail because of these changes. What is this technique? (1 Marks)

- A. Mutation testing
- B. Fuzz testing
- C. Regression testing
- D. Static analysis

Q10. During a software development lifecycle phase, the team conducts a formal walkthrough of the requirements specification document to identify ambiguities. What type of testing does this represent? (1 Marks)

- A. Dynamic testing
- B. Static testing
- C. Functional testing
- D. Performance testing

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Regression testing ensures that code changes have not adversely affected existing features, whereas re-testing specifically verifies the bug fix.

Q2. Correct Answer: A

Rationale: Boundary Value Analysis focuses on testing the exact boundary values and their immediate neighbors.

Q3. Correct Answer: B

Rationale: Three valid partitions exist (under 18, 18-60, and over 60), requiring at least one test case per partition.

Q4. Correct Answer: B

Rationale: Top-down integration testing uses stubs to simulate lower-level modules that are not yet integrated or developed.

Q5. Correct Answer: C

Rationale: Endurance testing evaluates system performance under a sustained continuous load over a long period.

Q6. Correct Answer: B

Rationale: White-box testing involves analyzing the internal structure, code, and logic of the software.

Q7. Correct Answer: B

Rationale: Beta testing is conducted by external users in real-world environments before the final commercial release.

Q8. Correct Answer: B

Rationale: Exploratory testing is an informal technique where test design and execution occur concurrently to maximize learning.

Q9. Correct Answer: A

Rationale: Mutation testing measures the effectiveness of a test suite by modifying the source code and checking if tests fail.

Q10. Correct Answer: B

Rationale: Static testing involves examining software work products, such as requirements or code, without executing the program.